

5) Data Collection and Initial Exploration.

```
[ ]  import pandas as pd
df = pd.read_csv("clean_sales_dataset(1).csv")

print(df)

[ ]

print("First 5 rows:")
print(df.head())

print("\nLast 5 rows:")
print(df.tail())

print("\nDataset Shape:")
print(df.shape)

print("\nColumn Names:")
print(df.columns)

print("\nDataset Information:")
print(df.info())

print("\nStatistical Summary:")
print(df.describe())

print("\nMissing Values:")
print(df.isnull().sum())

print("\nData Types:")
print(df.dtypes)
```

```

Last 5 rows:
...   SerialNo      Date      Product  Sales  Quantity  Region
20      21  2026-05-02      Laptop  88000         5  Hyderabad
21      22  2026-05-09  Mobile Phone  45000         9    Chennai
22      23  2026-05-15      Headphones  3800        16  Bangalore
23      24  2026-05-21      Keyboard  2600        20  Hyderabad
24      25  2026-05-28        Mouse  1350        25    Chennai

Dataset Shape:
(25, 6)

Column Names:
Index(['SerialNo', 'Date', 'Product', 'Sales', 'Quantity', 'Region'], dtype='object')

Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 25 entries, 0 to 24
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   SerialNo    25 non-null     int64
1   Date        25 non-null     object
2   Product     25 non-null     object
3   Sales       25 non-null     int64
4   Quantity    25 non-null     int64
5   Region      25 non-null     object
dtypes: int64(3), object(3)
memory usage: 1.3+ KB
None

Statistical Summary:
      SerialNo      Sales  Quantity
count  25.000000    25.00000  25.000000
mean   13.000000   24618.00000  12.000000
std     7.359801   31150.28384   6.601767
min     1.000000    900.00000   2.000000
25%     7.000000   2100.00000   6.000000
50%    13.000000   3200.00000  12.000000
75%    19.000000  41000.00000  16.000000
max    25.000000  88000.00000  25.000000

```

Missing Values:

```

SerialNo    0
Date        0
Product     0
Sales       0
Quantity    0
Region      0
dtype: int64

```

Data Types:

```

SerialNo    int64
Date        object
Product     object
Sales       int64
Quantity    int64
Region      object
dtype: object

```

```

import matplotlib.pyplot as plt
import seaborn as sns

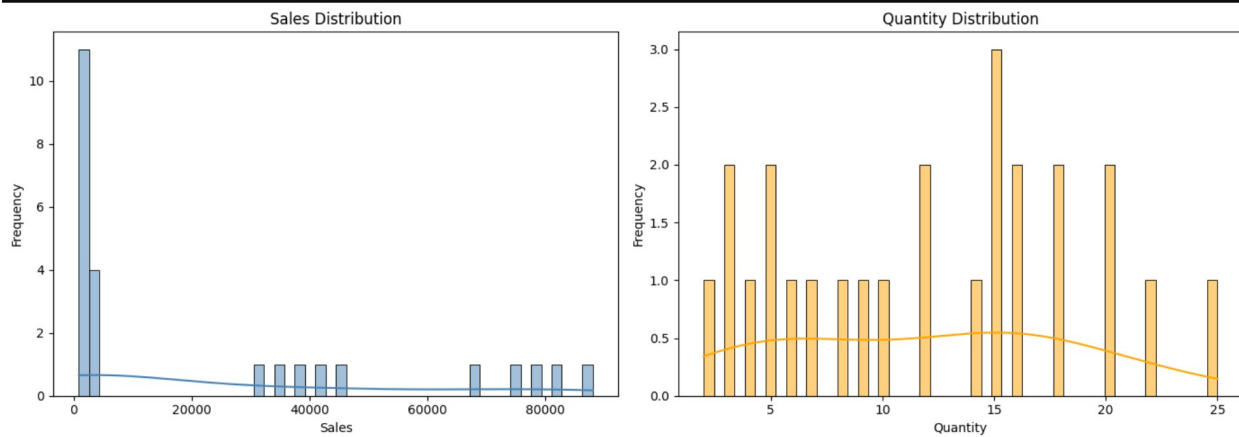
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# Sales histogram
sns.histplot(df['Sales'], bins=50, kde=True, color='steelblue', ax=axes[0])
axes[0].set_title('Sales Distribution')
axes[0].set_xlabel('Sales')
axes[0].set_ylabel('Frequency')

# Quantity histogram
sns.histplot(df['Quantity'], bins=50, kde=True, color='orange', ax=axes[1])
axes[1].set_title('Quantity Distribution')
axes[1].set_xlabel('Quantity')
axes[1].set_ylabel('Frequency')

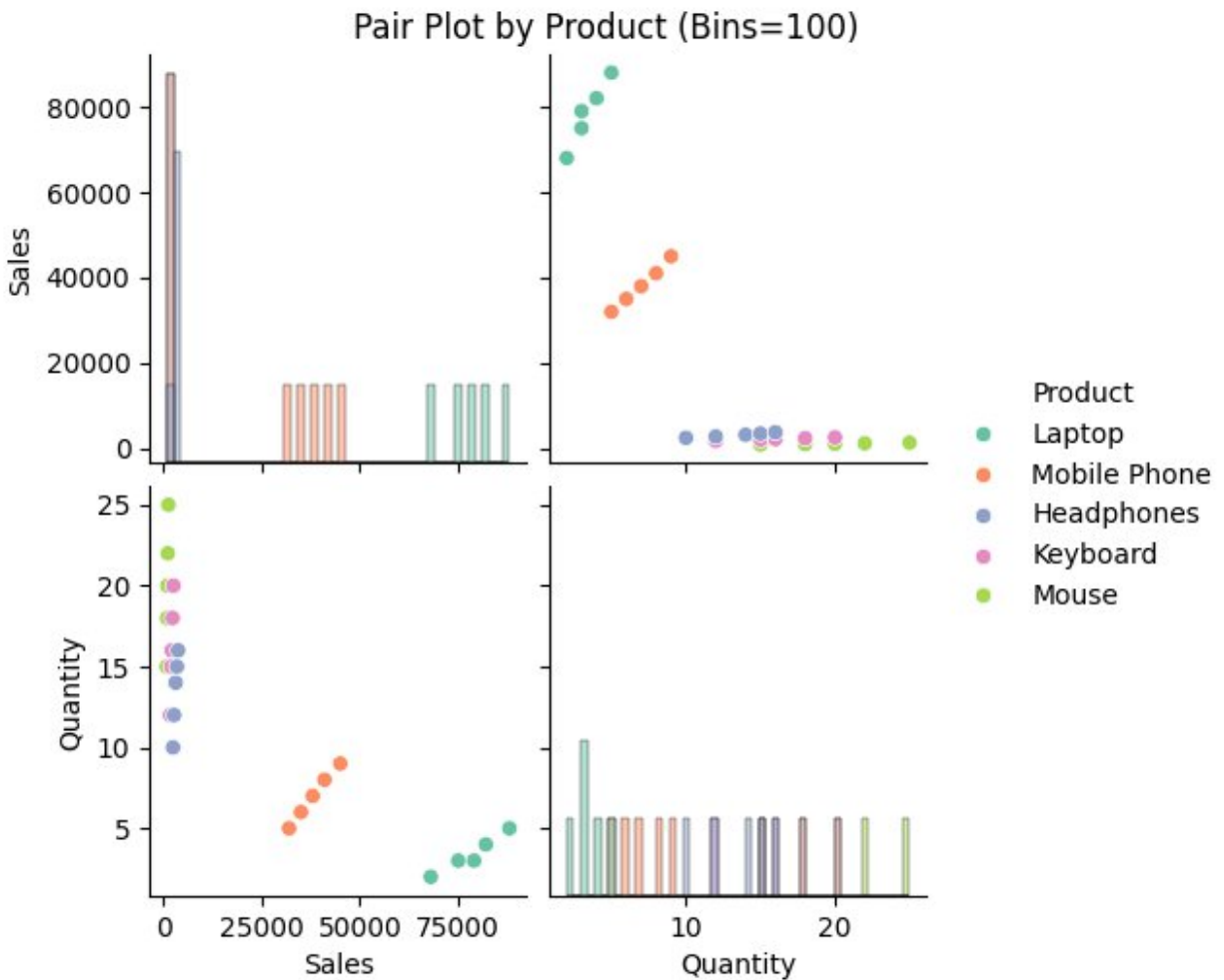
plt.tight_layout()
plt.show()

```



```
plt.figure(figsize=(10, 10))
sns.pairplot(df,
             vars=['Sales', 'Quantity'],
             hue='Product', # Color by Product category
             diag_kind='hist',
             diag_kws={'bins': 50},
             palette='Set2')

plt.suptitle("Pair Plot by Product (Bins=100)", y=1.02)
plt.show()
```



```
df.hist(bins = 50,figsize = (20,15))  
plt.show()  
sns.pairplot(df)  
plt.show()
```

